

A Note: The first posted version of this file was an abridged version - this is the original, to have been posted as 7 messages, with some corrections made for grammar and typeface.

June, 26, 2013

Bismillahi Ar-Rahmani Ar-Raheem

"The reward of deeds depends upon the intentions and every person will get the reward according to what he has intended."

*- Muhammad (S), the Messenger of Allah,
From the first Hadith of Sahih Bukhari*

To The Kryptos Yahoo Users Group,

I'm aware that many have been searching for a solution to the K4 section. However, have you ever wondered about K3. What is it about? How does it relate to the first two sections? It's about finding a long lost tomb.

What does that have to do with Intelligence, Communications, Information Gathering, or any other discipline of spycraft?

Maybe Nothing.

In the first two sections, K1 and K2, Jim Sanborn, Kryptos' creator, used a Vigenère cipher to encrypt their messages. Those required some amount of skill to decipher. However, the third section, K3, was encoded using letter transposition. Although they can be a bit maddening to work out, it was a much simpler technique.

A Vigenère cipher's weakness is correlated to the length and predictability of its key word. The longer and less predictable the key word, the harder it is to break the cipher. What would make the best key? It would be a key at least the same length as the message it encodes, and it would be chosen to be no more predictable than any other key its length - that is, chosen in a completely random way.

When using a key of this type, the cipher is known as a "one-time pad". As long as the key is kept secret by its users and only used to encrypt a single message, the ciphertext cannot be cracked.

What if K4 is encrypted using a one-time pad? As I remember, the builders of the pyramids of Ancient Egypt would build two chambers. One path, the direct one, would lead to a false chamber containing nothing of value as to deceive the tomb raiders. The other, off to the side from the direct one, would lead to the pharaoh's resting place - and to his treasure.

What if the decipherment of K3 is really a false chamber - and the key to K4 is hidden in the ciphertext of K3?

- Ibn Şifre

SUMHGGJ

part 1 of 7

Bismillahi Al Bari

*"I think we've got something, sir.
The report is only a fragment, from a probe droid in the Hoth System,
but it's the best lead we've had."*

*...
"You've found something?"
- "Yes M'Lord!"*

*Darth Vader, Capitan Peitt and Admiral Ozzel
The Empire Strikes Back*

To start, let's try and find BERLIN. BERLIN would imply a pad text of ELYOIE. Searching in the ciphertext of K3, Fig 1, we do not find it.

1 ENDYAHROHNLSRHEOCPTIOIBIDYSHNAIACHTNREYULDSLSSL
2 NOHSNOSMRWXMNETPRNGATIHNRARPESLNNELEBLPIACAEWMT
3 WNDITEENRAHCTENEUDRETNAEOETFOLSEDTIWENHAEIOYTEY
4 QHEENCTAYCREIFTBRSPAMHHEWENATAMATEGYEERLBTEEFOS
5 FIOTUETUAEOTOARMAERTNRTIBSEDDNIAAHTTMSTEWPIEROA

6 GRIWFEBAECTDDHILCEIHSITEGOEAOSSDRYDLORITRKLMLH
7 AGTDHARDPNEOHMGFMFEUHEECDMRIPFEIMEHNLSSSTRTVDOHW

Fig. 1

However, we do have something: EIOY backwards, as shown in Fig. 2, which is an interesting coincidence.

1 ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIACHTNREYULDSLSSL
2 NOHSNOSMRWXMETPRNGATIHNRRAPESLNNELEBLPIACAEWMT
3 WNDITEENRAHCTENEUDRETNHAEOTFOLSEDTIWENHAEIOYTEY
4 QHEENCTAYCREIFTBRSPAMHHEWENATAMATEGYEERLBTEEFOS
5 FIOTUETUAEOTOARMAEERTNRTIBSEDDNIAAHTTMSTEWPIEROA
6 GRIWFEBAECTDDHILCEIHSITEGOEAOSSDRYDLORITRKLMLH
7 AGTDHARDPNEOHMGFMFEUHEECDMRIPFEIMEHNLSSSTRTVDOHW

Fig. 2

And we do find, on row 6, LE, the two beginning letters. Also, let us assume that "level two" means two levels of encryption. The first being by a mask of some sort to find reveal the pad and the second the pad itself. Perhaps the mask goes something like shown in Fig. 3.

1 ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIACHTNREYULDSLSSL
2 NOHSNOSMRWXMETPRNGATIHNRRAPESLNNELEBLPIACAEWMT
3 WNDITEENRAHCTENEUDRETNHAEOTFOLSEDTIWENHAEIOYTEY
4 QHEENCTAYCREIFTBRSPAMHHEWENATAMATEGYEERLBTEEFOS
5 FIOTUETUAEOTOARMAEERTNRTIBSEDDNIAAHTTMSTEWPIEROA
6 GRIWFEBAECTDDHILCEIHSITEGOEAOSSDRYDLORITRKLMLH
7 AGTDHARDPNEOHMGFMFEUHEECDMRIPFEIMEHNLSSSTRTVDOHW

Fig. 3

That would give us: HELYOIETIROWITDDDSOFT... Using this as the pad and the proper letters of K4 as ciphertext, we work backwards to find the plain text, Fig. 4.

O B K R U O X O G H U L B S O L I F B B W F L R
V Q Q P R N G K S S O T W T Q S J Q S S E K Z Z
W A T J K L U D I A W I N F B N Y P V T T M Z E
P K W G D K Z X T J C D I G K U H U A U E K C A
R

Fig. 4

HELYOIETIROWITDDDS
BNYPVTMTZFPKWGDKZX

QBERLINHDEXPBCKJIM

Which in a less abbreviated English might mean:
? Berlin Head, expect back Jim

I believe this is interesting.

- Ibn Şifre

Note: I, hopefully, have posted or included a python script that describes the exact way that one calculates the plaintext from the pad and ciphertext.

KKIPETC

part 2 of 7

Bismillahi Al Muhsi

"Turning the accomplishment of many years. Into an hour-glass:"
Prologue, Henry V, Shakespeare

Or, in this case, even less than that. Continuing along the same path, come to a message in such a way:

There are only 97 letters of K4. If K3 is a pad, all the the letters will not be used. First let us take the last twenty

four columns of the K3 ciphertext and then reverse their order such as in Fig. 5.

	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W
1	L	L	S	L	S	D	L	U	Y	E	R	N	T	H	C	A	I	A	N	H	S	Y	D	
2	T	M	W	E	A	C	A	I	I	P	L	B	E	L	E	N	N	L	S	E	P	R	A	R
3	Y	E	T	Y	O	I	E	A	H	N	E	W	I	T	D	E	S	L	O	F	T	E	O	E
4	S	A	O	F	E	E	T	B	L	R	E	E	Y	G	E	T	A	M	A	T	A	N	E	W
5	A	O	R	E	I	P	W	E	T	S	M	T	T	H	A	A	I	N	D	D	E	S	B	I
6	H	E	L	M	L	K	R	T	I	R	O	L	D	Y	R	D	D	S	O	A	E	O	G	E
7	W	H	O	D	V	T	R	T	T	S	S	L	N	H	E	M	I	E	F	P	I	R	M	D

Fig. 5

Then adjust the rows as shown in Fig. 6a and then 6b

	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W		
1	L	L	S					L	U	Y	E							C	A	I	A			D		
2	T	M	W					I	I	P	L							N	N	L	S			R		
3	Y	E	T					A	H	N	E							E	S	L	O			E		
4	S	A	O	L	L	S	D	B	L	R	E	R	N	T	H			T	A	M	A	N	H	S	Y	W
5	A	O	R	E	A	C	A	E	T	S	M	B	E	L	E	A	I	N	D	E	P	R	A	I		
6	H	E	L	Y	O	I	E	T	I	R	O	W	I	T	D	D	D	S	O	F	T	E	O	E		
7	W	H	O	F	E	E	T	T	T	S	S	E	Y	G	E	M	I	E	F	T	A	N	E	D		
1				E	I	P	W					T	T	H	A					D	E	S	B			
2				M	L	K	R					L	D	Y	R					A	E	O	G			
3				D	V	T	R					L	N	H	E					P	I	R	M			

Fig. 6a

	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W
1	L	L	S	E	I	P	W	L	U	Y	E	T	T	H	A	C	A	I	A	D	E	S	B	D
2	T	M	W	M	L	K	R	I	I	P	L	L	D	Y	R	N	N	L	S	A	E	O	G	R
3	Y	E	T	D	V	T	R	A	H	N	E	L	N	H	E	E	S	L	O	P	I	R	M	E
4	S	A	O	L	L	S	D	B	L	R	E	R	N	T	H	T	A	M	A	N	H	S	Y	W
5	A	O	R	E	A	C	A	E	T	S	M	B	E	L	E	A	I	N	D	E	P	R	A	I
6	H	E	L	Y	O	I	E	T	I	R	O	W	I	T	D	D	D	S	O	F	T	E	O	E
7	W	H	O	F	E	E	T	T	T	S	S	E	Y	G	E	M	I	E	F	T	A	N	E	D

Fig. 6b

Now if we match the K4 ciphertext with the modified K3, used as a pad, as shown in Fig. 7a, 7b.

	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4
	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W
1	L	L	S	E	I	P	W	L	U	Y	E	T	T	H	A	C	A	I	A	D	E	S	B	D
2	T	M	W	M	L	K	R	I	I	P	L	L	D	Y	R	N	N	L	S	A	E	O	G	R
3	Y	E	T	D	V	T	R	A	H	N	E	L	N	H	E	E	S	L	O	P	I	R	M	E
4	S	A	O	L	L	S	D	B	L	R	E	R	N	T	H	T	A	M	A	N	H	S	Y	W
5	A	O	R	E	A	C	A	E	T	S	M	B	E	L	E	A	I	N	D	E	P	R	A	I
6	H	E	L	Y	O	I	E	T	I	R	O	W	I	T	D	D	D	S	O	F	T	E	O	E
7	W	H	O	F	E	E	T	T	T	S	S	E	Y	G	E	M	I	E	F	T	A	N	E	D

Fig. 7a

	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1
1	O	B	K	R	U	O	X	O	G	H	U	L	B	S							
2	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O
3	T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W							
5	A	T	J	K	L	U	D	I	A	W	I	N	F	B							
6	N	Y	P	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D
7	I	G	K	U	H	U	A	U	E	K	C	A	R								

Fig. 7b

We get the following plaintexts:

Row 1:

OBKRUXOGHULBS
UYETHACIADES

DSIWLLLVSHAWK

Row 2:

OLIFBBWFLRVQPRNGKSSO
WMLKRIIPLLDYRNLSAEOG

BZXFANBCKDFMNDBYANURM

Row 3:

TWTQSQSSEKZZW
ELNHEESLOPIRME

NSESUOHIRBEXAF

Row 5:

ATJKLUDIAWINFB
CAETSMBELEAIND

XWOVEPYTJFBTNX

Row 6:

NYPVTMZFPKWGDZXTJCD
ELYOIETIROWITDDDSOFTE

BERLINHDEXPBCKJIMZTOZ

Row 7:

IGKUHUAUEKCAR
TTSSEYGEMIEFT

ECQIPNQDNEXUW

Consider that the plaintext of rows 3 and 7 are reversed and we get the following message:

DSIWLLLVSHAWK
BZXFANBCKDFMNDBYANURM
FAXEBRIHOUSESN

XWOVEPYTJFBTNX
BERLINHDEXPBCKJIMZTOZ
WUXENDQNPIQCE

TWGTGGP

- Ibn Şifre

part 3 of 7

Bismillahi Al Latif

*"For it is better for a scribe to be thought ignorant than to pay the penalty for the detection of plans",
De Furtivis Literarum Notis,
Giovanni Battista Porta*

*A quote on Porta's "suggesting the deliberate misspellings of plaintext words", taken from
The Codebreakers,
David Khan*

Each plaintext of K1, K2 and K3 contained a misspelling for a total of three misspellings. However, if we are working on the assumption that the plaintext of K3 was misleading then we have only two real misspellings. If let us assume that each row contains three - or maybe two - misspellings.

We could construct the following message in Fig. 9:

```

  S   L   Z
DCIWILLVSAHAWK

  Z       F       U
BRXFANBCKDEMNDBYANORM

  X   I   U
FAKEBRCHOOSESNS

  W   E   J
XSOVSPYTOFBTNX

                Z   Z
BERLINHDEXPBCKJIMSTOP
  U       Q       Q
WNXENDINPIECE

```

Fig. 9

Which would give:

```

1 DCI WILL VS A HAWK
2 BR X FAN BACK DEMND BY ANORM
3 FAKE BR CHOOSE SN
4
5 X SOV SPY TO FB TNX
6 BERLIN HD EXP BCK JIM STOP
7 WN X END IN PIECE

```

Or perhaps, in a little better English, something like the following

DCI Will [Director of Central Intelligence William (Webster)] vs a hawk brother.

Fan back: demand by abnormal fake brother, choose soon. Soviet spy to fake brother: thanks Berlin head, expect back - Jim stop win. End in Piece.

Now, one can have a lot of fun coming up with messages by using these techniques, but this possible solution is interesting for the misspellings.

Observe, and note L and O are the correct letters for the misspellings of K1 and K2 plaintexts respectively:

Put the misspelled letters in order of appearance as seen in Fig. 10 (for rows 3 and 7 that means reversed from before)

```

K1 K2  1   2   3   5   6   7
L   O   CIA REO OCK SSO SP EIN

LOCI ARE OOCKS SO SPE IN

          O           E
LOCI ARE ROCKS SO SPY IN

```

Fig. 10

Loci are rocks so spy in ...

Now this might refer to the plaintext of K2 and the rocks in Kryptos' sculpture. There is one more curious observation about this particular message. If we swap the rows we get a somewhat different message.

Observe Fig. 11:

```

1 DCI WILL VS A HAWK
5 X SOV SPY TO FB TNX
2 BR X FAN BACK DEMND BY ANORM
6 BERLIN HD EXP BCK JIM STOP
3 FAKE BR CHOOSE SN
7 WN X END IN PIECE

```

Fig. 11

Or perhaps, again, in a little better English:

DCI Will [Director of Central Intelligence William (Webster)] vs a hawk.

Soviet spy to fake brother: thanks brother. Fan back: demand by a normal Berlin head, expect back - Jim stop fake brother - choose soon - win! End in Piece.

In this scenario, the misspellings become as shown in Fig. 12:

```

K1,Q K2,U 1 5 2 6 3 7
L O CIA SSO REO SP OCK EIN
LO CIA SSORE 0 SPOCK EIN
S C E
LO CIA SCORE 0 SPOCK WIN

```

Fig. 12

Lo, CIA score 0, Spook win.

Which is ominous (assuming you're not the spy)! It appears that the plaintext read one way might mean that the spy is caught, or in another way the spy wins.

IBZJEJN

- Ibn Şifre

part 4 of 7

Bismillahi Al Musawwir

*"Once is happenstance. Twice is coincidence. Three times, it's enemy action."
Auric Goldfinger
You Only Live Twice*

It would be a shame to waste those letters from the misspellings. And notice too that there is a gaping hole in this guess. Namely row four of the part of the rearranged K3 we have been using:

The misspellings:

SLZ ZFU UIX WEJ ZZ QQU

Row 4 of Fig 7a

SAOLLSDBLRERNHTAMANSY

```

SLZZFUUIXWEJZZQQU
LSDBLRERNHTAMANH
-----
ITIEILUQDHONWFMAGRA

```

or reversed:

ARGAMFWNOHDQULIEI

AR GAMF WN O HD Q U LIE I

A E N

BR GAME WW O HD Q U LIE I

Again perhaps, in a little better English, something like the following:

Brothers Game. William Webster - O, Head? You Lie!

And these new misspellings: WEB (remember BEW is in reverse order), so that might end the misspelled message as:

LOCI ARE ROCKS SO SPY IN WEB

Spy's caught! - Well maybe.

Oh, I almost forgot, about the alignment of K3' and K4. If you use the Kryptos alphabet to label the columns as in Fig. 7 one gets at the start of each mapping:

R1: b
R2: y
R3: d (left side of mapping start)

R5: o
R6: r
R7: a (left side of mapping start)

Perhaps:

BY D OR A (down or across)?

Or perhaps:

BY DORA?

Maybe it's in reference to the Morse code at the edge of one of the rocks?

If you want to look at the lettering of the next column over for each mapping, using the other row order:

R1: c
R5: s
R2: p
R6: y
R3: w (right side of mapping end)
R7: n (right side of mapping end)

Perhaps:

C SPY WN (See spy win)

What has come before this is the best solution that I have found, but consider ...

JKKQRXW

- Ibn Şifre

part 5 of 7

Bismillahi Al Alim

*"But they were all of them deceived, for another ring was made.",
Galadriel,*

Opening narration of The Lord of the Rings: The Fellowship of the Ring (film)

One thing about using a one-time pad; once you've used it - never, never, never, NEVER EVER use it again.

Did Jim use it again? The way that the rearranged K3, which I'll call K3', and K4 line up in Fig.7 seems a bit arbitrary. There is more that can be found using these techniques and different alignments of K3' and K4. Whether or not it is coincidence, I'm not sure.

For instance Fig. 13a with figure 13b

	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W
1	L	L	S	E	I	P	W	L	U	Y	E	T	T	H	A	C	A	I	A	D	E	S	B	D
2	T	M	W	M	L	K	R	I	I	P	L	L	D	Y	R	N	N	L	S	A	E	O	G	R
3	Y	E	T	D	V	T	R	A	H	N	E	L	N	H	E	E	S	L	O	P	I	R	M	E
4	S	A	O	L	L	S	D	B	L	R	E	R	N	T	H	T	A	M	A	N	H	S	Y	W
5	A	O	R	E	A	C	A	E	T	S	M	B	E	L	E	A	I	N	D	E	P	R	A	I
6	H	E	L	Y	O	I	E	T	I	R	O	W	I	T	D	D	D	S	O	F	T	E	O	E
7	W	H	O	F	E	E	T	T	T	S	S	E	Y	G	E	M	I	E	F	T	A	N	E	D

Fig. 13a

	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1
1	O	B	K	R	U	O	X														
2	O	G	H	U	L	B	S	O	L	I	F	B	B	W							
3	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O							
4	T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W							
5	A	T	J	K	L	U	D	I	A	W	I	N	F	B							
6	N	Y	P	V	T	T	M	Z	F	P	K	W	G	D							
7	K	Z	X	T	J	C	D	I	G	K	U	H	U	A							
8	U	E	K	C	A	R															

Fig. 13b:

So briefly perhaps:

OFEETTSSEYGEM
 KZXTJCDIGKUHA

 UGGNFOSCAINRDI

Or: UG **IN** **POS** **MAIN** RD I

LLSEIPWLUYETHH
 NYPVTMZFPKWGD

 YEWEIRUBLRINCV

Or: **N** CNTR **L** BURIED **EY**

RERNHTTAMANHSY
 ATJKLUDIAWINFB

 SNIAGASBIJVOSS

Or: **ES** **ON** **JIM** S AGAIN S

MBELEAINDEPRAI
 OGHULBSOLIFBBW

 GOPTSRLFATCARE

Or: BR ACT**S** **F**IRST PO**I**

OIETIRO
 OBKRUOX

 KNIWSTN

Or: NTS WIN**S**

Perhaps - LAST PIG, ITS OINK!

Are these part of the first solution presented, is it a mix of the two, or is all of this completely wrong - or just partly right?

CCYXFSB

- Ibn Şifre

part 6 of 7

Bismillahi Al Haqq il Ala

"Waterhouse opens his mouth and says 'Gxnn bhldh sqrd m!' " in Outer Qwghlm to Mary

... Waterhouse is then punched in the face by Mary's date ...

"With us [from Inner Qwghlm] ... what you just said isn't a polite greeting" to Waterhouse from the RAAF chap

" Oh ... what did I say, then?"

"You said that while you were down at the mill to lodge a complaint about a sack with a weak seam that sprung loose on Thursday, you were led to understand, by the tone of the proprietor's voice, that Mary's great-aunt, a spinster who had a loose reputation as a younger woman, had contracted a fungal infection in her toenails."

- RAAF chap

"No, no! ... I understood him to say that it was at the pub, and that it was my neighbor's dog that had come down with rabies" - Mary

"He was at the basilica for confession--the priest--angina--" - someone else

"The dockside--Mary's half-sister--leprosy--Wednesday--complaining about a loud party" - everyone else at once

"I was trying to say that she looked beautiful"

- Waterhouse

"Ah! ... Then you should have said 'Gxnn bhldh sqrd m' "

- Rod speaking in Inner Qwghlm

*Dialog from the
Cryptonomicon,
Neal Stephenson*

After a while, or perhaps much sooner than that, one can't help but question this solution - is it right? To quote a great Sage, "I may be as woefully wrong as Humphrey Belcher, who believed the time was ripe for a cheese cauldron", but it's the best that I have found to date – and I'm not sure that I like it. I've written this in hopes that this might help someone from this group, or someone else, to find the solution. If you follow the idea that the encryption of K4 is a pad revealed by a mask of K3, this still leaves a lot of play to be had, as seen. But before you go to Jim and ask - wait, just wait. And Think. Think about it. Write your idea down and put it away somewhere for a week, or a month, or a year – maybe more. Then, dust it off and look at it again. I've been prejudiced by this type solution, and hope someone this group can help take it further, or debunk it with something better.

One thing about the misspellings - my, my the misspellings; one can go on almost forever constructing new messages with misspellings. I recommend to anyone who wants to 'play' in this way, to become both a believer and a disbeliever in coincidence. Which leaves me thinking; using them in this way with broken, condensed English, seems to allow one to be able to hide a message in a set of believable, valid, but different messages. That's not a bad idea as far as cryptography goes - if you can pull it off.

So let's solve this, and figure out what that solution means.

*Salam and happy hunting,
Ibn Şifre*